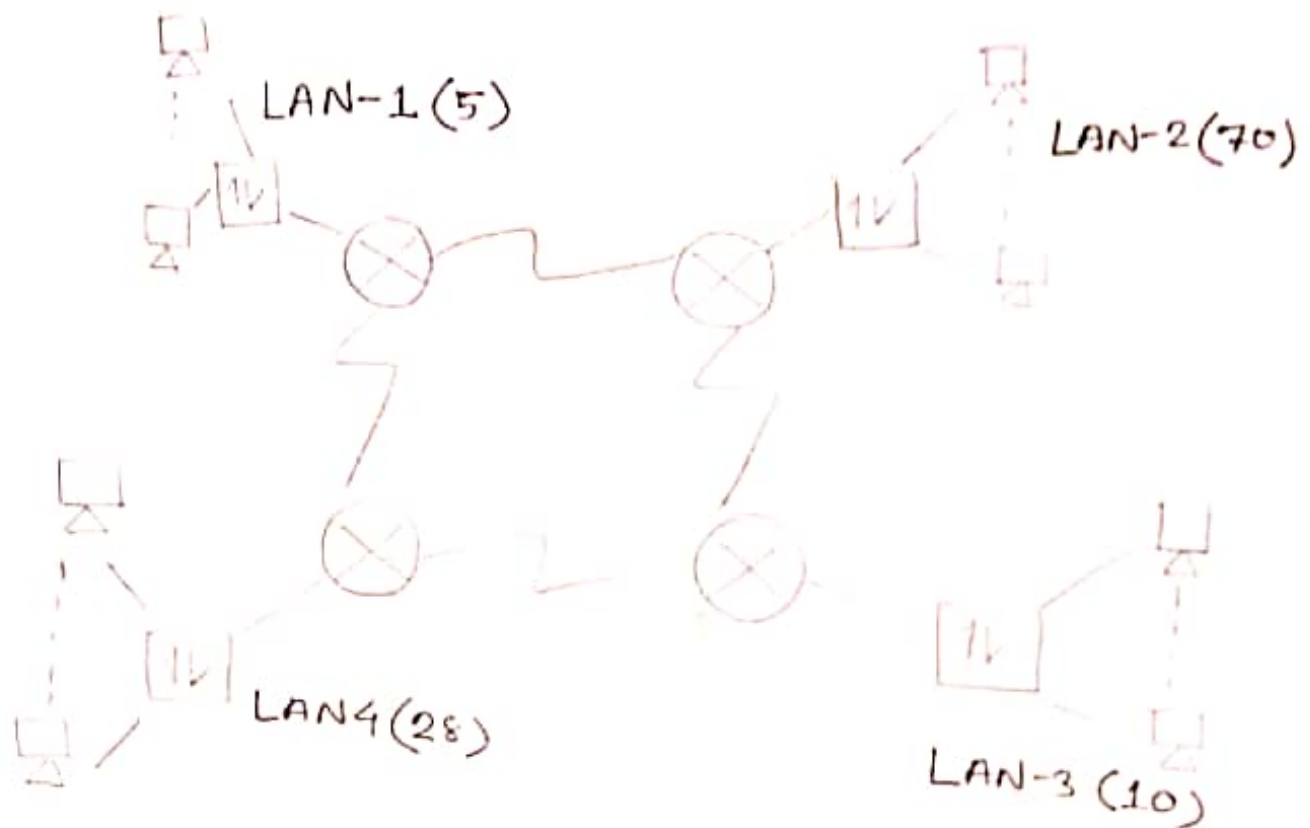


Given Network:-



Here are number of 8 Networks:-

1. LAN 1: 5 hosts.
 2. LAN 2: 70 hosts
 3. LAN 3: 10 hosts
 4. LAN 4: 28 hosts.
- and 4 serial Network of 2 hosts each.

Given Base Address, 200.200.200.0/24

Maximum number of hosts is 70.

to facilitate 70 hosts we need 7 bits for hosts as $2^7 - 2 = 126 > 70$.

Base address 200.200.200.0/24

Possible subnets with 7 bits for hosts

1) 200.200.200. 00000000 /25 $\rightarrow$ 200.200.200.0/25

2) 200.200.200. 1 00000000 /25 $\rightarrow$ 200.200.200.128/25
7 bit for host

the ip 200.200.200.0/25 is assigned to LAN-2.

Now we'll subnet remaining 200.200.200.128/25 for LAN-4 having 28 hosts.

We'll need $(2^5 - 2 > 28)$ so 5 bits for host portion.

$\therefore$ possible subnets;

- 1) 200.200.200.128/27 (LAN-4)
- 2) 200.200.200.160/27
- 3) 200.200.200.192/27
- 4) 200.200.200.224/27

As,

$$(8 - 5) + 24 = 27.$$

27 - 25 = 2 bits for combination

that means $2^2 = 4$ subnets.

Now, 200.200.200.160/27 will be used to create subnets.

For LAN-3 We have 10 hosts.

$\therefore$ 4 host bits required

$\therefore$ possible subnets:-

- 1) 200.200.200.160/28 (LAN-3)
- 2) 200.200.200.176/28

Now,

200.200.200.176/28 will be used for subnetting.

For LAN-1 there are 5 hosts so 3 bits are required.

∴ possible subnets:-

1) 200.200.200.176/29 [LAN-1]

2) 200.200.200.184/29

Now, remaining 200.200.200.184/29 can be used for subnetting:-

required hosts per subnet 2 and number of subnets 4.

So, 2 bits for 2 hosts, ∴

1) 200.200.200.184/30 [serial-1]

2) 200.200.200.188/30 [serial-2]

2 subnets more required so we can use the unused 200.200.200.192/27 for that as base address.

∴ 1) 200.200.200.192/30 and [serial-3]

2) 200.200.200.196/30 [serial-4].

* Given Lan number and number of hosts under each Lan:-

LAN-1	10 hosts
LAN-2	80 hosts
LAN-3	1000 hosts
LAN-4	3000 hosts
LAN-5	20 hosts
LAN-6	1500 hosts
LAN-7	2 hosts
LAN-8	2 hosts

and base address, 175.170.32.0/19

To facilitate 3000 hosts we need 12 host bits as $2^{12} - 2 = 4096 - 2 = 4094$

∴ possible subnets:-

- 1) 175.170.32.0/20 (assigned to LAN-4)
- 2) 175.170.48.0/20

Now base address will be considered as 175.170.48.0/20

for 1500 hosts we need 11 host bits

∴ possible subnets:-

- 1) 175.170.48.0/21 (assigned to LAN-6)
- 2) 175.170.56.0/21

Now new base address 175.170.56.0/21

Number of hosts 1000

bits required 10 for hosts

∴ possible subnets:-

1) 175.170.56.0/22 (Assigned to LAN-3)

2) 175.170.60.0/22

New base address considered 175.170.60.0/22

Number of hosts 80

bits required 7 for hosts

∴ possible subnets:-

1. 175.170.60.0/25 (Assigned to LAN-2)

2. 175.170.60.128/25

3. 175.170.64.0/25

:

Now, taking 175.170.60.128/25 as base address

number of hosts 20.

number of host bits required = 5

∴ possible subnets:-

1) 175.170.60.128/27 assigned to LAN-5

2) 175.170.60.160/27

3) 175.170.60.192/27

4) 175.170.60.224/27

Now, remaining network address 175.170.60.160/27

number of hosts 10.

number of bits required for host is 4.

∴ possible subnets:-

1) 175.170.60.160/28 (for LAN-1)

2) 175.170.60.176/28

Now for LAN-7 and LAN-8 we need 2 subnets each having 2 hosts.

∴ number of host bits required 2.

∴ possible subnets:-

1) 175.170.60.176/30 (LAN-7)

2) 175.170.60.180/30 (LAN-8)

3) 175.170.60.184/30

4) 175.170.60.188/30

* Given IP Address 172.16.0.0/16.

To create 100 subnets we require 7 bits extra for network.

∴ 1st subnet: 172.16.0.0/23

1. total Subnet = $2^7 = 128$
2. New Subnet Mask 255.255.255.0
3. Hosts/per subnet = $2^{32-23} - 2 = 510$
4. total hosts = $128 \times 510 = 65280$

Number of hosts if not been subnetting,

$$2^{32-16} - 2 = 65534$$

$$\therefore \text{hosts lost} = 65534 - 65280 = 254.$$

* Given IP Address 192.168.2.0/24
12 subnets to be created.

For this we require 4 network bits extra.

∴ 1st subnet address: 192.168.2.0/28

1. total Subnet = $2^4 = 16$.
2. New Subnet Mask 255.255.255.240
3. Host per subnet = $2^{32-28} - 2 = 14$
- 4.

$$\text{total hosts} = 14 \times 16 = 224$$

$$\text{total hosts before subnetting} = 2^8 - 2 = 254$$

$$\text{hosts lost for subnetting} = 254 - 224 = 30.$$